



Al-Kufa University
Engineering College



Electronic And Communication Department

3rd stage

LABORATORY MANUAL

Sequential Logic Circuit Lab

"Laboratory Experiments of Digital Logic Circuits"

Experiment 1

Logic Behavior of 7408 & 4013 IC Gates

Prepared by:
Asst.lec. Muhiialdin M.ridha

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Doings

- ✓ Maintain discipline & silence.
- ✓ Keep the Laboratory clean and tidy.
- ✓ Handle instruments with utmost care.
- ✓ **Come prepared with laboratory manual each experiment and pen.**
- ✓ Return all the issued equipment properly.
- ✓ Get the signature on experiment result sheet before you leave.
- ✓ For any clarification contact faculty/staff in charge only.
- ✓ Shut down the power supply after the experiment and return all the equipment to its proper place.

Attentions:

- ✗ Avoid unnecessary chat or walk.
- ✗ Do not start the experiment without instructions.
- ✗ Avoid using cell phones unless absolutely necessary.
- ✗ **Avoid touching the legs of the IC's in your bare hand.**

Aim: To get familiar with the use of different logic gates IC's.

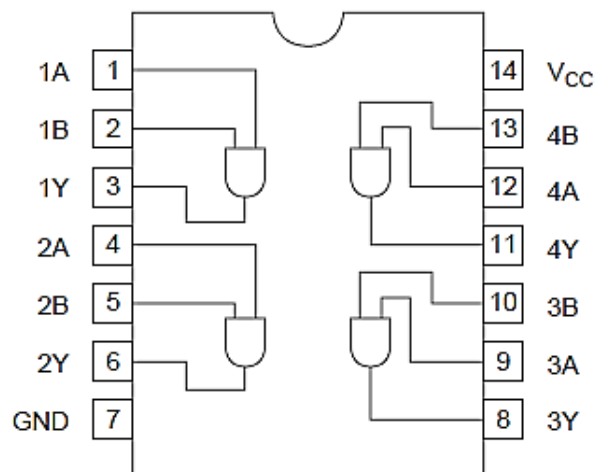
In this experiment, you will investigate the logic behavior of various IC gates:

- **HD74LS08P Quadruple 2-Input Positive AND Gates**
- **CD4013BE CMOS Dual D Flip Flop**

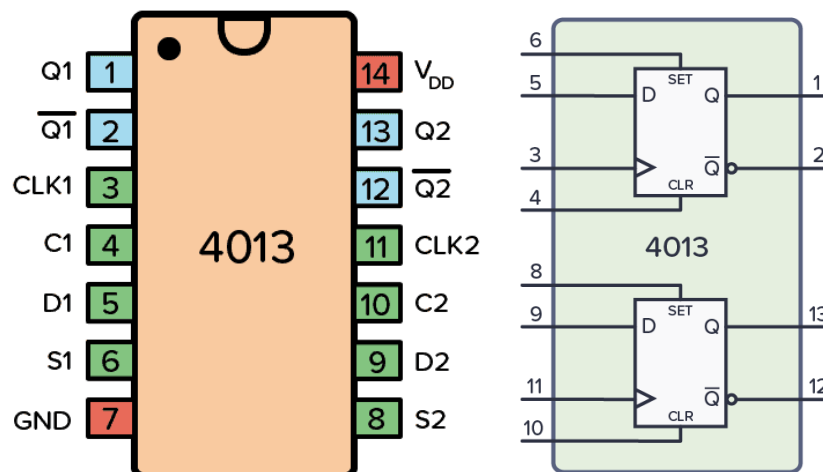
Theory :

In this experiment, you will design, construct, and test one combinational logic circuit and one sequential logic circuit.

1) HD74LS08P



2) CD4013BE



Procedure

1. Use one gate from each IC mentioned in the theory and obtain the truth table of the gate. The truth table is obtained by connecting the inputs of the gate to switches and the output to an indicator LED.
2. Use whatever resistors and capacitors you need in your circuit design.
3. Draw the circuit diagram clarifying the connection of the switches and the resistors for the both 2 IC's mentioned



Circuit Diagram of HD74LS08P



Circuit Diagram of CD4013BE

Discussions

1. Observe the datasheet of the IC's mentioned in this experiment then define the following:

V_{iL} : I_{iL} :

V_{iH} : I_{iH} :

V_{oL} : I_{oL} :

V_{oH} : I_{oH} :

2. Based on the experiment you made what is the triggering type of CD4013BE IC?
3. What is the advantage of edge triggering over level triggering?